



ArthaRoshni

The 500-Lab Economics Skills Catalogue

Free technical labs, notes & research skills for
Class 10 · 11 · 12 and early college — built for every student

by **Roshni**

A practical curriculum you can build online for free — using Google Colab, open data (World Bank, RBI, MoSPI), and open-source tools. Inspired by the technical facilities of the world's leading economics departments, redesigned so any student can access them at zero cost.

Generated 12 July 2026 · 500 labs · 10 skill tracks

1 · Vision & Who This Is For

ArthaRoshni is a free, offline-first learning platform for economics students in Classes 10–12 and early college — with a special focus on students in India (and Bihar) who rarely get exposure to the global-level "technical labs" that top universities run. Those labs (econometrics computing, experimental economics, data-science) usually need expensive software and infrastructure. This catalogue re-creates their **spirit** using free, open tools so cost is never a barrier.

What a student gets: clear notes, PPT/reading material, quizzes, and 500 structured hands-on labs that teach real research skills — Python, statistics, econometrics, NLP/sentiment analysis, behavioural experiments, climate and development economics, and a capstone research track.

Honest scope note. The 500 labs below are ArthaRoshni's own curriculum — activities you can build and run for free. Section 2 separately summarises what real top universities offer (with sources); it does not claim these 500 are named foreign-university labs.

2 · What the World's Top Universities Offer

Leading universities increasingly blend economics with computing, data science and AI. A few real, verifiable examples of the programs, tools and skills they teach — which this catalogue mirrors with free tools:

Institution / Program	Focus & tools	Skills taught
MIT — BSc Computer Science, Economics & Data Science (Course 6-14)	Microeconomic theory, econometrics, algorithms, optimization, machine learning	Game theory, modelling, data analytics
ETH Zurich — CAS in AI, Data & Machine Learning	Python, SQL, data science & ML end-to-end, AI in industry	Programming, data, ML pipelines
UC Berkeley — Econometrics Laboratory (EML)	Stata, R, EViews, RATS, MATLAB on a shared computing facility	Applied econometrics & research computing
UC Santa Barbara — Experimental & Behavioral Economics Lab (EBEL); UC Berkeley XLab	oTree, z-Tree experiment software	Designing & running economic experiments
University of Zurich — MSc Informatics: Artificial Intelligence	90 ECTS with minors incl. economics & computational linguistics	AI, data science, NLP
Text-as-data in economics research	Loughran–McDonald finance dictionary; Economic Policy Uncertainty index; VADER/NLTK	Sentiment & tone of policy/news text
Indian institutes (IITs / IIMs / ISI / Delhi School of Economics)	Econometrics, data science, financial economics with Python/R/Stata	Quantitative & applied economics

Sources: MIT Course Catalog (6-14); ETH Zurich School for Continuing Education (CAS AI, Data & ML); UC Berkeley Econometrics Laboratory (eml.berkeley.edu); UC Santa Barbara EBEL; University of Zurich MSc AI; Loughran–McDonald master dictionary (sraf.nd.edu); policyuncertainty.com.

3 · AI, Robotics & Emerging Techniques in Economics

Modern economics is being reshaped by computation. The labs in this catalogue introduce students to:

- **Python & R** for data handling, econometrics and reproducible research (free via Google Colab / RStudio).
- **Machine learning** for prediction and causal inference (Lasso, random forests, double ML).
- **NLP & sentiment analysis** — measuring the tone of RBI statements, budgets and news ("text-as-data").
- **Behavioural & experimental economics** — online games (oTree) to study real decision-making.
- **Climate & environmental economics** — emissions data, carbon pricing, cost–benefit analysis.
- **Computational & agent-based modelling** — simulating markets and policy.
- **AI/robotics link** — automation's effect on labour markets, algorithmic pricing, and AI for economic forecasting.

4 · How to Build These Labs Online — For Free

The whole stack costs **■0**. Requirements and steps:

Layer	Free tool	Purpose
Compute / notebooks	Google Colab	Run Python labs in the browser, no install
Language & libraries	Python, pandas, NumPy, matplotlib, statsmodels, scikit-learn, NLTK	Data, stats, econometrics, NLP
Statistics (alt.)	R + RStudio, Gretl	Free econometrics packages
Experiments	oTree, Google Forms	Behavioural games & surveys
Open data	World Bank, RBI, MoSPI, IPCC, Our World in Data, UN Comtrade	Real datasets for every lab
Website & hosting	Cloudflare Pages + GitHub	Publish the app free with a free domain
Database / admin	Supabase (free tier)	Store notes, PPTs, users (optional)

Steps to launch one lab: (1) write it as a Colab notebook with objective + data link + code + a quiz; (2) add a page on the website that embeds/links the notebook; (3) attach the dataset and a short assessment; (4) publish. Repeat per lab — the table below gives you all 500 to build.

5 · The 500-Lab Master Catalogue

10 tracks × 50 labs. Columns: ID · Title · Level · Key skills · Tools · Output. Build each as a free notebook or web activity.

■ Track 1: Python Foundations for Economics

50 labs · Sample / open CSV · Google Colab, Python, pandas

ID	Lab title	Level	Key skills	Output
001	Open Colab & run your first economics notebook	10–12	Notebook use	Working notebook
002	Store income, price & GDP in variables	10–12	Variables, types	Calculations
003	Arithmetic with economic values	10–12	Operators	Result table
004	Handle dataset labels & units	10–12	Strings	Labeled output
005	Store a year of monthly prices	10–12	Lists, indexing	Price list
006	Fix economic constants (tax rate, base year)	10–12	Tuples	Constant set
007	Map states to indicators	10–12	Dicts	State lookup
008	Classify inflation as low/high	10–12	if/else	Decision logic
009	Compute growth for many years	10–12	for loops	Repeated output
010	Build a reusable CAGR calculator	10–12	Functions	Custom function
011	Take a class survey value	10–12	input(), casting	User entry
012	Test poverty-line thresholds	10–12	Comparisons	True/False flags
013	Fix common Python errors	10–12	Error reading	Corrected script
014	Document your economics code	10–12	Comments	Annotated notebook
015	Open & save a text/CSV file	10–12	File I/O	Saved file
016	Percentage change of a price series	10–12	Percent math	Growth values
017	Per-capita income from totals	10–12	Ratios	Per-capita table
018	Future value of savings	10–12	Compound interest	Value table
019	Loan interest calculator	10–12	SI formula	Interest output
020	Monthly EMI estimator	11–12	Loan math	EMI figure
021	Simulate dice/lottery payoffs	10–12	random module	Simulated data
022	Generate a year sequence	10–12	range()	Year list
023	Sort states by GDP	10–12	sorted()	Ranking
024	Unique product categories	10–12	Sets	Category set
025	Round & format currency	10–12	f-strings	Clean output
026	Build a price×quantity grid	11–12	Nested loops	Revenue grid
027	Quick growth-rate list	11–12	Comprehensions	Compact code
028	Loop over state indicators	11–12	Dict loops	Summary print
029	Guard against bad input	11–12	try/except	Robust script
030	Import math for economics	10–12	Imports	Reusable tools
031	Arrays for fast calculation	11–12	NumPy	Array output
032	Vectorised price changes	11–12	Vectorisation	Fast results
033	Aggregate a data array	10–12	NumPy stats	Aggregates
034	Organise data into a matrix	11–12	Reshaping	Matrix
035	Flag above-average incomes	11–12	Boolean masks	Flags
036	Elasticity calculator	11–12	Formula coding	Elasticity value
037	Break-even point tool	11–12	Cost/revenue	Break-even units
038	Compute GDP = C+I+G+NX	11–12	Identity coding	GDP figure
039	Build a simple price index	11–12	Index math	Index series
040	Deflate nominal GDP	11–12	Deflator	Real GDP
041	Years-to-target savings	10–12	Loops	Timeline
042	Income-tax slab calculator	11–12	Conditions	Tax due
043	Household budget allocator	10–12	Dicts	Budget split
044	Convert ■ ↔ \$ with a rate	10–12	Arithmetic	Converted value

ID	Lab title	Level	Key skills	Output
045	Project population growth	11–12	Compounding	Projection
046	Build a small dataset by code	10–12	Data structures	Dataset
047	One-line quick plot	10–12	matplotlib	Chart
048	Save & re-run a notebook	11–12	Docs, versions	Reproducible file
049	Handle survey data responsibly	10–12	Data ethics	Checklist
050	Your first Python economics project	10–12	Full workflow	Starter project

■ Track 2: Data Handling & Visualization

50 labs · World Bank / MoSPI CSV · Colab, pandas, matplotlib

ID	Lab title	Level	Key skills	Output
051	Read a CSV into pandas	10–12	read_csv	Imported data
052	head(), info(), describe()	10–12	Inspection	Data summary
053	Detect missing values	10–12	isna()	Missing report
054	Fill or drop missing data	10–12	fillna/dropna	Clean dataset
055	Standardise column names	10–12	rename()	Tidy columns
056	Pick indicators with loc/iloc	10–12	Selection	Subset
057	Filter by year or state	10–12	Boolean filter	Filtered view
058	Rank states by GDP	10–12	sort_values	Ranking
059	Join two datasets by key	11–12	merge()	Joined file
060	Stack yearly files	11–12	concat()	Combined data
061	Average by group	11–12	groupby	Group summary
062	Category comparison table	11–12	pivot_table	Pivot
063	Custom column transforms	11–12	apply()	Derived column
064	Parse year/date fields	11–12	datetime	Date-ready data
065	Categories to codes	11–12	map/replace	Recoded vars
066	Group incomes into bands	11–12	cut()	Income bands
067	Remove duplicate rows	10–12	drop_duplicates	Clean rows
068	Save a cleaned CSV	10–12	to_csv	Exported file
069	Plot GDP over time	10–12	matplotlib	Trend chart
070	Compare state indicators	10–12	bar plot	Bar chart
071	Compare two years by state	11–12	grouped bars	Comparison
072	Income vs education	11–12	scatter	Relationship
073	Income distribution	11–12	hist	Distribution
074	Compare group spreads	11–12	boxplot	Box chart
075	Sector shares of GDP	10–12	pie	Share chart
076	Stacked sector output	11–12	area plot	Area chart
077	Compare two series	11–12	multi-line	Dual plot
078	Dashboard of 4 charts	11–12	subplots	Figure
079	Label key points on a chart	10–12	annotations	Clear chart
080	Colours, theme & fonts	10–12	styling	Polished chart
081	Correlation heatmap	11–12	heatmap	Correlation map
082	Plot fast-growing series	11–12	log axis	Log chart
083	Inflation & rates together	12+	twin axis	Dual-axis chart
084	Export PNG/SVG	10–12	savefig	Downloadable figure
085	Hoverable chart (plotly)	12+	plotly	Interactive plot
086	Colour states by value	12+	geo plot	Choropleth
087	Moving-average trend	11–12	rolling	Smoothed series
088	Cumulative growth curve	11–12	cumsum	Cumulative chart
089	Index a series to 100	11–12	normalisation	Indexed series
090	Spot & mark outliers	11–12	IQR rule	Outlier chart
091	Wide ↔ long (melt/pivot)	12+	melt	Reshaped data
092	Attach indicator to map	12+	geo merge	Mapped data
093	One-page indicator board	12+	layout	Dashboard
094	Explain a dataset in words	10–12	narrative	Data story
095	Spot misleading charts	10–12	critique	Error analysis
096	Compare growth paths	11–12	compare	Comparison chart
097	State development dashboard	11–12	regional viz	State board
098	Turn data into a poster	11–12	design	Infographic
099	Flag unreliable values	11–12	validation	Quality report
100	Data visualization capstone	11–12	full workflow	Chart story

■ Track 3: Statistics & Probability

50 labs · Survey / open data · Colab, pandas, NumPy

ID	Lab title	Level	Key skills	Output
101	Average income lab	10–12	Mean	Mean value
102	Median vs mean	10–12	Median	Comparison
103	Most common category	10–12	Mode	Result
104	Measure spread	10–12	Range	Spread value
105	Variability of marks	11–12	Variance	Variance
106	Dispersion of incomes	11–12	Std dev	Spread report
107	Income percentiles	11–12	Percentiles	Percentile table
108	Quartiles & IQR	11–12	Quartiles	Quartile summary
109	Standardise values	11–12	Z-scores	Standardised data
110	Shape of income data	12+	Skew	Shape note
111	Detect extreme values	11–12	IQR	Outlier list
112	Build a frequency table	10–12	Tallies	Frequency table
113	Two-category comparison	11–12	crosstab	Contingency table
114	Basic probability cases	10–12	Probability	Answers
115	Independent events	11–12	Event logic	Table
116	Conditional probability	11–12	Bayes basics	Result
117	Update beliefs with evidence	12+	Bayes rule	Posterior
118	EV of a lottery	11–12	EV	EV value
119	Normal distribution intro	12+	Normal	Bell curve
120	Coin/trial simulation	11–12	Binomial	Probability plot
121	Draw random samples	11–12	Sampling	Sample compare
122	Understand bias	11–12	Bias	Discussion
123	Sampling distribution of mean	12+	CLT	Sampling plot
124	Standard error of mean	12+	SE	SE value
125	CI for a mean	12+	Inference	CI output
126	Compare two means	12+	t-test	Test output
127	Test category association	12+	chi ²	Test result
128	Compare 3+ group means	12+	ANOVA	F-test
129	Correlation coefficient	11–12	Pearson r	Correlation value
130	Spearman rank correlation	12+	Spearman	Rank r
131	Correlation ≠ causation	11–12	Reasoning	Concept note
132	Simple & weighted index	11–12	Index	Index series
133	Build a mini CPI	11–12	Price index	CPI value
134	Inflation from CPI	11–12	Rate math	Inflation report
135	Compute growth rates	10–12	Percent change	Growth table
136	Gini coefficient of income	12+	Inequality	Gini value
137	Lorenz curve lab	12+	Inequality	Lorenz curve
138	Head-count poverty ratio	11–12	Poverty math	Ratio
139	Weighted average price	11–12	Weights	Weighted value
140	Smoothing a noisy series	11–12	Rolling mean	Smoothed data
141	Build a good questionnaire	10–12	Survey design	Instrument
142	Consent & privacy	10–12	Ethics	Checklist
143	Monte-Carlo dice economy	12+	Simulation	Distribution
144	Randomly assign treatment	12+	RCT basics	Assignment
145	Compare two policies	12+	A/B test	Comparison
146	Estimate elasticity from data	12+	Elasticity	Estimate
147	Naive forecast baseline	11–12	Forecast	Prediction
148	Read uncertainty	11–12	Interpretation	Written note
149	Statistics & probability quiz	10–12	Recall	Quiz score
150	Class survey full analysis	10–12	Full workflow	Short report

■ Track 4: Econometrics I

50 labs · World Bank / RBI series · Colab, statsmodels

ID	Lab title	Level	Key skills	Output
151	Correlation vs causation	11–12	Interpretation	Concept note
152	Fit a line by eye vs OLS	11–12	Line fitting	Comparison
153	Income vs education years	11–12	OLS	Regression output
154	Interpret the slope	11–12	Coefficients	Plain result
155	Meaning of the intercept	11–12	Interpretation	Note
156	Model fit explained	11–12	R ²	Fit note
157	Predict from the model	11–12	Fitted values	Prediction
158	Residual plot basics	12+	Residuals	Residual plot
159	Growth vs unemployment	12+	Applied OLS	Okun estimate
160	Inflation vs unemployment	12+	Applied OLS	Phillips estimate
161	Estimate price–quantity	11–12	Demand	Slope estimate
162	Income vs food share	12+	Engel	Curve estimate
163	Score ~ hours + income	12+	Multiple OLS	Model output
164	Why add control variables	12+	Confounders	Comparison
165	Gender/region dummy variable	12+	Dummies	Dummy model
166	Interaction terms	12+	Interactions	Interaction result
167	Log wage regression	12+	Logs	Semi-elasticity
168	Constant-elasticity model	12+	Elasticity	Elasticity
169	Diminishing returns curve	12+	Nonlinear	Curve plot
170	Read standard errors	12+	SE	SE note
171	t-statistics of coefficients	12+	Testing	Interpretation
172	Significance of an effect	12+	p-values	p-value note
173	CI for a coefficient	12+	Inference	CI table
174	Joint significance	12+	F-test	Test result
175	Adjusted R ² & comparison	12+	Model choice	Comparison
176	VIF & correlated Xs	12+	VIF	Diagnostic
177	Non-constant variance	12+	BP test	Diagnostic plot
178	Heteroskedasticity-robust SE	12+	Robust SE	Corrected output
179	Omitted-variable bias demo	12+	Bias	Comparison
180	Choosing the right form	12+	Specification	Best model
181	Influential points	12+	Leverage	Influence check
182	Standardised coefficients	12+	Beta coeffs	Ranked effects
183	Price elasticity of demand	12+	Elasticity	Estimate
184	Estimate a wage gap	12+	Applied OLS	Gap estimate
185	Mincer earnings model	12+	Mincer	Return estimate
186	Estimate MPC	12+	Consumption fn	MPC
187	Investment vs interest rate	12+	Applied OLS	Estimate
188	Money demand estimate	12+	Applied OLS	Estimate
189	Cross-country growth	12+	Applied OLS	Estimate
190	Drivers of poverty	12+	Applied OLS	Model report
191	Income vs life expectancy	11–12	Applied OLS	Estimate
192	Rainfall vs crop yield	11–12	Applied OLS	Estimate
193	In-sample vs out-of-sample	12+	Validation	Error compare
194	Avoid the dummy trap	12+	Encoding	Fixed model
195	Write a regression result	11–12	Writing	Result paragraph
196	Replicate a published result	12+	Replication	Replicated output
197	Honest reporting of results	11–12	Ethics	Checklist
198	statsmodels vs sklearn	12+	Tooling	Same result twice
199	Regression concepts quiz	11–12	Recall	Quiz score
200	Econometrics I mini-paper	12+	Full workflow	Short paper

■ Track 5: Econometrics II (Advanced)

50 labs · Panel / time-series data · Colab, statsmodels, linearmodels

ID	Lab title	Level	Key skills	Output
201	Plot & decompose a series	12+	Decomposition	Trend/seasonal
202	ADF unit-root test	12+	Stationarity	Test result
203	Make a series stationary	12+	Differencing	Stationary series
204	ACF & PACF plots	12+	ACF/PACF	Correlogram
205	Autoregressive model	12+	AR	Fitted model
206	Moving-average model	12+	MA	Fitted model
207	ARIMA forecast of inflation	12+	ARIMA	Forecast
208	Seasonal ARIMA	12+	Seasonality	Forecast
209	RMSE / MAE of forecasts	12+	Error metrics	Accuracy
210	HP filter trend & cycle	12+	Filtering	Cycle plot
211	Rolling volatility	12+	Volatility	Risk series
212	Volatility clustering test	12+	ARCH	Test result
213	GARCH volatility model	12+	GARCH	Volatility forecast
214	Engle–Granger test	12+	Cointegration	Test result
215	Error-correction model	12+	ECM	Model output
216	Two-variable VAR	12+	VAR	System output
217	Granger-causality test	12+	Causality	Test result
218	Impulse-response function	12+	IRF	Response plot
219	Long vs wide panel data	12+	Panel setup	Structured data
220	Pooled panel regression	12+	Pooled OLS	Estimate
221	Within (FE) estimator	12+	FE	FE output
222	RE estimator	12+	RE	RE output
223	FE vs RE choice	12+	Hausman	Decision
224	Cluster-robust errors	12+	Clustering	Corrected SE
225	Policy DiD evaluation	12+	DiD	Treatment effect
226	Event-study design	12+	Event study	Effect path
227	Endogeneity problem	12+	IV logic	Concept
228	Two-stage least squares	12+	2SLS	IV estimate
229	First-stage F-check	12+	IV diagnostics	Strength check
230	Regression discontinuity	12+	RDD	Local effect
231	Propensity-score matching	12+	Matching	Balanced sample
232	Binary outcome (logit)	12+	Logit	Probabilities
233	Probit model	12+	Probit	Marginal effects
234	Interpret nonlinear effects	12+	MXF	Effect table
235	Choice among options	12+	Multinomial	Choice probs
236	Poisson regression	12+	Poisson	Rate model
237	Quantile regression	12+	Quantiles	Distributional effect
238	Two-way fixed effects	12+	2FE	Output
239	Seasonal adjustment	12+	Adjustment	Adjusted series
240	Nowcast GDP from indicators	12+	Nowcasting	Estimate
241	Compare prediction vs inference	12+	ML/econ	Comparison
242	Lasso variable selection	12+	Regularisation	Selected vars
243	Tree model for prediction	12+	ML	Predictions
244	k-fold validation	12+	CV	Validated error
245	Double ML intro	12+	Causal ML	Effect estimate
246	Bootstrap standard errors	12+	Resampling	Bootstrap SE
247	Sensitivity of results	12+	Robustness	Robustness table
248	Publication-quality tables	12+	Reporting	Formatted table
249	Replicate a macro study	12+	Replication	Replicated result
250	Advanced econometrics project	12+	Full workflow	Research note

■ Track 6: Behavioral & Experimental Economics

50 labs · Class-collected responses · Google Forms / oTree, Colab

ID	Lab title	Level	Key skills	Output
251	Dictator game (■100 split)	10–12	Social preferences	Giving data
252	Ultimatum bargaining game	10–12	Fairness	Offer/accept data
253	Trust & reciprocity game	11–12	Trust	Transfer data
254	Contribution to a common pot	11–12	Cooperation	Contribution data
255	Prisoner's dilemma	10–12	Game theory	Choice data
256	Anchoring on a number	10–12	Cognitive bias	Estimate compare
257	Gain vs loss framing	10–12	Framing	Choice compare
258	Endowment/loss-aversion demo	11–12	Loss aversion	Response data
259	Choose safe vs risky	11–12	Risk preference	Risk estimate
260	■100 now vs ■120 later	11–12	Discounting	Discount rate
261	Opt-in vs opt-out saving	11–12	Nudge	Take-up data
262	Money labelled by purpose	11–12	Mental accounting	Choice data
263	Judging likelihood from memory	10–12	Heuristics	Comparison
264	Seeking agreeing evidence	11–12	Bias	Reflection
265	Sunk-cost fallacy test	11–12	Bias	Choice data
266	Calibration quiz	11–12	Overconfidence	Calibration
267	Do it now vs later	11–12	Present bias	Choice data
268	Design a savings nudge	11–12	Nudge design	Nudge plan
269	Guess ■ of the average	11–12	Strategic thinking	Number data
270	First-price sealed-bid auction	12+	Auctions	Bid data
271	Vickrey auction demo	12+	Auctions	Bid data
272	Asset-market bubble game	12+	Markets	Price path
273	Coordination bank-run game	12+	Coordination	Outcome data
274	Fair-wage survey	10–12	Fairness	Survey data
275	Dictator-to-charity variant	11–12	Altruism	Giving data
276	Gift-exchange game	12+	Reciprocity	Effort data
277	Norm-elicitation survey	12+	Norms	Norm ratings
278	Repeated public goods	12+	Dynamics	Trend data
279	Costly punishment variant	12+	Enforcement	Behaviour data
280	Compare risk by group	11–12	Comparison	Group data
281	Willingness-to-accept vs pay	11–12	Endowment effect	Gap data
282	Belief in streaks	11–12	Bias	Belief data
283	Independent events belief	11–12	Bias	Response data
284	Scarcity & decision quality	11–12	Scarcity	Comparison
285	Too many options test	11–12	Choice overload	Choice data
286	Prime then decide	12+	Priming	Comparison
287	Status-quo bias test	11–12	Bias	Choice data
288	Decisions with peer info	12+	Peer effects	Behaviour data
289	Die-roll honesty task	12+	Dishonesty	Reported data
290	Chat then choose	12+	Communication	Outcome data
291	Histogram of dictator offers	10–12	Analysis	Distribution
292	Ultimatum offer analysis	10–12	Analysis	Chart
293	Fit risk aversion	12+	Estimation	Parameter
294	Fit a discount rate	12+	Estimation	Parameter
295	Behaviour vs rational model	11–12	Comparison	Written note
296	Set up a simple oTree app	12+	oTree	Running app
297	Google Form → Colab analysis	10–12	Pipeline	Analysis
298	Design a new experiment	12+	Design	Protocol
299	Experiment consent & debrief	11–12	Ethics	Consent form
300	Behavioural economics report	11–12	Full workflow	Lab report

■ ■ Track 7: NLP & Text Analytics

50 labs · News / policy text · Colab, NLTK / VADER

ID	Lab title	Level	Key skills	Output
301	Load an economics text file	11–12	File I/O	Loaded text
302	Lowercase & strip punctuation	11–12	Cleaning	Clean text
303	Split text into words	11–12	Tokenisation	Tokens
304	Remove common words	11–12	Stopwords	Filtered tokens
305	Word-frequency count	11–12	Counting	Frequency table
306	Visualise frequent words	10–12	Word cloud	Cloud image
307	Document-term matrix	12+	BoW	Matrix
308	Weight important words	12+	TF-IDF	Weighted terms
309	Score tone by dictionary	11–12	Lexicon	Tone score
310	Sentiment with VADER	12+	VADER	Sentiment scores
311	Loughran–McDonald finance tone	12+	Finance NLP	Tone
312	Sentiment over time	12+	Time NLP	Trend chart
313	Analyse a Union Budget speech	11–12	Applied NLP	Findings
314	Central-bank statement tone	12+	Applied NLP	Tone score
315	Classify headline sentiment	11–12	Classification	Labels
316	Basic Hindi text handling	11–12	Multilingual	Processed text
317	Compare Hindi vs English tone	12+	Comparison	Comparison
318	Common two-word phrases	12+	n-grams	Phrase list
319	Words that go together	12+	Collocations	Pairs
320	Extract firms & places	12+	NER	Entity list
321	LDA topics in reports	12+	LDA	Topics
322	Top keywords of a report	12+	Keywords	Keyword list
323	Count uncertainty words (EPU)	12+	Text index	Index value
324	Compare two documents	12+	Cosine similarity	Similarity score
325	Extractive summary	12+	Summarisation	Summary
326	Train a news classifier	12+	ML text	Model
327	NB sentiment classifier	12+	Naive Bayes	Predictions
328	Logistic-regression on text	12+	ML text	Predictions
329	Split & evaluate a model	12+	Validation	Accuracy
330	Confusion matrix read	12+	Evaluation	Matrix
331	Word vectors intro	12+	Embeddings	Vectors
332	Responsible text collection	11–12	Ethics	Guidelines
333	Sentiment of economic posts	12+	Social NLP	Sentiment
334	Consumer-review sentiment	11–12	Applied NLP	Sentiment
335	10-K/annual-report tone	12+	Finance NLP	Tone
336	Party manifesto economics words	12+	Text analysis	Findings
337	Track a keyword over years	12+	Trend NLP	Trend
338	Beyond positive/negative	12+	Emotion NLP	Emotion mix
339	Readability of policy text	11–12	Readability	Score
340	Tone across newspapers	12+	Comparison	Comparison
341	Spot exaggerated claims	12+	Critical NLP	Assessment
342	Sentiment dashboard	12+	Dashboard	Dashboard
343	End-to-end text pipeline	12+	Pipeline	Reusable pipeline
344	Use an LLM to label text	12+	LLM basics	Labels
345	Prompt design for extraction	12+	Prompting	Structured output
346	Bias in sentiment tools	12+	Fairness	Reflection
347	Replicate a text-as-data study	12+	Replication	Result
348	Write up a text finding	11–12	Writing	Report
349	NLP concepts quiz	11–12	Recall	Quiz score
350	Economic sentiment capstone	12+	Full workflow	Research note

■ Track 8: Climate & Environmental Economics

50 labs · World Bank / IPCC / Our World in Data · Colab, pandas

ID	Lab title	Level	Key skills	Output
351	Load global CO ₂ emissions	11–12	Open data	Dataset
352	CO ₂ per capita by country	11–12	Ratios	Comparison
353	India–China–USA emissions	11–12	Comparison	Chart
354	Emissions trend over time	11–12	Time series	Trend chart
355	Coal/renewables share	11–12	Shares	Mix chart
356	Growth of solar & wind	11–12	Growth	Chart
357	Emissions per unit GDP	12+	Intensity	Intensity series
358	Growth vs emissions	12+	Analysis	Decoupling plot
359	Environmental Kuznets curve	12+	Applied model	Curve
360	Model a carbon tax on prices	12+	Policy model	Price effect
361	Social cost of carbon intro	12+	Valuation	Estimate
362	CBA of a green project	12+	CBA	Net benefit
363	Discounting the future	12+	Discounting	NPV
364	Temperature → damage	12+	Modelling	Damage curve
365	AQI & economic cost	11–12	Applied data	Cost estimate
366	Water stress indicators	11–12	Open data	Analysis
367	Rainfall & crop yield	11–12	Applied data	Relationship
368	Economic loss from disasters	11–12	Data	Loss chart
369	Heat vs labour productivity	12+	Applied data	Estimate
370	Climate & migration link	12+	Analysis	Findings
371	Forest cover change	11–12	Data	Trend
372	Access to electricity	11–12	Development	Analysis
373	Fossil-fuel subsidy data	12+	Policy data	Analysis
374	Emissions-trading price data	12+	Markets	Price chart
375	Falling cost of solar	11–12	Trend	Learning curve
376	Electric-vehicle uptake	11–12	Adoption	Curve
377	Peak demand & renewables	12+	Data	Analysis
378	Jobs in clean energy	11–12	Labour data	Estimate
379	Recycling & waste economics	11–12	Data	Analysis
380	Personal footprint calculator	10–12	Calculator	Footprint
381	Indian state emissions	12+	Regional data	Comparison
382	Bihar climate vulnerability	11–12	Regional	Profile
383	Track SDG climate targets	11–12	SDG data	Scorecard
384	Cost of adaptation	12+	CBA	Estimate
385	Adjust GDP for environment	12+	Accounting	Adjusted GDP
386	Green behaviour nudges	11–12	Behaviour	Plan
387	Pigouvian tax logic	12+	Theory+data	Illustration
388	Compare cap-and-trade vs tax	12+	Comparison	Note
389	Model reaching a RE target	12+	Projection	Path
390	Forecast future emissions	12+	Forecast	Projection
391	Who emits, who suffers	12+	Equity	Analysis
392	Green-finance flows	12+	Finance data	Analysis
393	Climate risk & insurance	12+	Risk	Analysis
394	Climate dashboard	11–12	Dashboard	Dashboard
395	Explain a climate dataset	10–12	Narrative	Data story
396	Write a climate policy brief	12+	Writing	Brief
397	Cost of net-zero for India	12+	Analysis	Argument
398	Replicate an emissions chart	11–12	Replication	Chart
399	Climate economics quiz	11–12	Recall	Quiz score
400	Climate economics capstone	12+	Full workflow	Research note

■ Track 9: International & Development Economics

50 labs · World Bank / UN Comtrade / WDI · Colab, pandas

ID	Lab title	Level	Key skills	Output
401	Load export/import data	11–12	Open data	Dataset
402	Compute the trade balance	11–12	Balance	Balance chart
403	India's top trade partners	11–12	Ranking	Chart
404	What India exports	11–12	Composition	Basket chart
405	What India imports	11–12	Composition	Basket chart
406	RCA index lab	12+	RCA	Index
407	Terms-of-trade series	12+	ToT	Series
408	Model a tariff's price effect	12+	Policy model	Effect
409	Quota vs tariff comparison	12+	Trade policy	Comparison
410	Gravity model of trade	12+	Applied OLS	Estimate
411	Exchange-rate trend	11–12	Time series	Chart
412	Real vs nominal FX	12+	REER	Series
413	Purchasing-power parity check	12+	PPP	Comparison
414	Big-Mac index lab	11–12	PPP	Valuation
415	Balance-of-payments structure	12+	BoP	Breakdown
416	FDI inflows analysis	11–12	Investment	Trend
417	Remittance flows to India	11–12	Flows	Chart
418	Forex reserves trend	11–12	Data	Chart
419	Current-account deficit	12+	BoP	Analysis
420	Effect of a weaker rupee	12+	Analysis	Note
421	Openness vs growth	12+	Applied OLS	Estimate
422	GVC participation	12+	GVC	Analysis
423	Tariff data across countries	12+	Policy data	Comparison
424	Trade effect of sanctions	12+	Analysis	Findings
425	Resource curse illustration	12+	Model+data	Illustration
426	Global poverty trends	11–12	Development	Trend
427	Inequality across countries	12+	Inequality	Comparison
428	Human Development Index lab	11–12	Composite index	HDI compare
429	Multidimensional poverty	12+	Index	Score
430	Do poor countries catch up?	12+	Growth	Test
431	Foreign-aid effectiveness	12+	Analysis	Findings
432	Microfinance outcomes	12+	Development	Analysis
433	Remittances & household welfare	12+	Applied	Estimate
434	Size of the informal economy	12+	Labour	Estimate
435	Female labour-force participation	12+	Gender	Analysis
436	Cross-country schooling	11–12	Comparison	Chart
437	Preston curve	12+	Applied	Curve
438	Urbanisation & growth	11–12	Analysis	Chart
439	Age structure & growth	12+	Demography	Analysis
440	Internal migration in India	12+	Development	Analysis
441	Track SDGs for India	11–12	SDG data	Dashboard
442	HDI across Indian states	12+	Regional	Comparison
443	Bihar vs India indicators	11–12	Regional	Profile
444	Estimate a poverty line	12+	Poverty math	Line
445	Effect of a transfer scheme	12+	Simulation	Effect
446	Simulate opening to trade	12+	Simulation	Outcome
447	A country's growth miracle	11–12	Case study	Note
448	Write a trade policy brief	12+	Writing	Brief
449	International & development quiz	11–12	Recall	Quiz score
450	Development economics capstone	12+	Full workflow	Research note

■ Track 10: Research, Policy & Capstone

50 labs · Student-chosen open data · Colab, docs, slides

ID	Lab title	Level	Key skills	Output
451	Frame a research question	11–12	Research design	Question
452	State a testable hypothesis	11–12	Hypotheses	Hypothesis
453	Mini literature review	12+	Reading, synthesis	Review
454	Define variables & measures	11–12	Operationalisation	Variable list
455	Plan data collection	11–12	Data strategy	Plan
456	Locate open datasets	11–12	Data search	Sources
457	Prepare data for analysis	11–12	Cleaning	Clean data
458	Exploratory data analysis	11–12	EDA	EDA report
459	Pick the right method	12+	Methodology	Method note
460	Run the main analysis	12+	Analysis	Results
461	Make publication charts	11–12	Visualisation	Figures
462	Interpret the findings	11–12	Interpretation	Discussion
463	State limitations honestly	11–12	Critical thinking	Limitations
464	Check robustness	12+	Robustness	Checks
465	Connect to a real policy	11–12	Policy analysis	Link
466	CBA of your proposal	12+	CBA	Net benefit
467	Research ethics review	11–12	Ethics	Checklist
468	Cite sources correctly	11–12	Citation	Bibliography
469	Write a 150-word abstract	11–12	Academic writing	Abstract
470	Write the full report	12+	Writing	Report
471	Design a research poster	11–12	Design	Poster
472	Build a presentation	10–12	Slides	Deck
473	Present & defend findings	11–12	Communication	Talk
474	Review a peer's project	12+	Peer review	Feedback
475	Revise from feedback	11–12	Revision	v2
476	Make it reproducible	12+	Reproducibility	Repo
477	Publish to the web	11–12	Publishing	Live page
478	Tell a data story	10–12	Storytelling	Story
479	Write an economics op-ed	11–12	Writing	Op-ed
480	Prepare a policy debate	11–12	Argumentation	Brief
481	Design & run a survey	11–12	Survey	Dataset
482	Small local field study	12+	Fieldwork	Findings
483	Local market price study	10–12	Fieldwork	Analysis
484	Analyse a govt budget	12+	Public finance	Analysis
485	Evaluate a welfare scheme	12+	Evaluation	Assessment
486	Business model economics	11–12	Applied	Model
487	Personal-finance capstone	10–12	Finance	Plan
488	Study a Bihar-specific issue	11–12	Regional	Report
489	Contribute to an SDG target	11–12	SDG	Project
490	Use AI responsibly in a project	12+	AI literacy	Reflection
491	Build a project portfolio	11–12	Portfolio	Portfolio
492	Collaborative research	11–12	Teamwork	Group report
493	Get mentor feedback	12+	Mentoring	Feedback
494	Prepare for an econ olympiad	12+	Problem solving	Entry
495	Map economics careers & skills	11–12	Career	Roadmap
496	Plan for higher studies	12+	Guidance	Plan
497	Free global courses list	11–12	Self-learning	List
498	Showcase day submission	11–12	Presentation	Submission
499	Reflect on what you learned	10–12	Reflection	Journal
500	Grand capstone research project	12+	Everything	Final paper

6 · Deploy & Scale

This catalogue is the content backbone of the ArthaRoshni web app. To publish it for free: push the project to GitHub, connect Cloudflare Pages (free hosting + a free *.pages.dev domain), and optionally connect Supabase for the admin panel. Full instructions are in **DEPLOYMENT.md**.

To expand beyond 500, break any lab into micro-labs (e.g. "simple regression" → load data, visualise, correlate, fit OLS, read residuals, interpret, write up). Regenerate this document any time with **python tools/build-pdf.py**.

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